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Question 1

1.1) 5 main benefits of adopting an agile approach;

-**Increase focus on business outcomes**, an agile approach encourages less well-defined documents upfront and more focus on taking an adaptive approach to deliver optimal value, teams are allowed to make required changes and be more creative and adaptive in their approach to achieve a maximized business value.

-**Reduced time to market**, by delivering value incrementally as soon as a deployable chunk of work has been developed/tested and approved by customers it can be deployed and this provides customers with the benefit of reduced time to market/access value.

- **Higher productivity and lower costs**, With Agile, we develop and test continuously within the same iteration. Teams resolve bugs/issues immediately, delivering a better outcome and reducing costs, rather than dealing with them later and risking customer dissatisfaction.

- **Higher quality**, in Agile approach, teams jointly take the shared responsibility in delivering project value optimally and individuals/teams are handed the control /resources /trust to optimize their input required to deliver project optimally this encourages teams to collaborate, be more effective /hands on and ensure misalignment do not lead to customer dissatisfaction which can be costly.

- **Organizational effectiveness**, with this approach a sense of control/organization is elaborated between teams they are motivated and effectively work together to deliver value and prioritize the growth and partnership in delivering value it becomes part of their lifestyle

1.2) how benefits can help address the challenges faced by Cisco Solutions;

-**Increase focus on business outcome**, with less focus on documentation and strict guidelines upfront, Cisco solutions can prioritize value driven techniques and become more adaptive/creative in delivering objectives in a manner that customer satisfaction is always achieved

With reduced time to market, Cisco solutions customers can access value as soon as a deployable chunk of work is completed. They can assess the product, test it in the market, receive feedback, and request changes if needed, ensuring maximized customer satisfaction.

High quality delivery, Cisco Solutions collaborates with customers, receiving continuous feedback (e.g., user personas). This feedback is incorporated incrementally, ensuring customer satisfaction and optimal value.

With higher productivity and lower costs, Cisco's team shares responsibility for delivering business value, boosting morale, and encouraging collaboration. This leads to effective and quick delivery of sprints, ensuring deadlines are met.

Organizational effectiveness, Continuous feedback/retrospectives help teams review and improve processes, eliminating inefficiencies and overlaps, and focusing on high-impact activities that enhance customer satisfaction.

Question 2

2.1) Stages of maturity in developing openness

Forming, at this stage the team is welcomed to a new collaborative environment. The team is polite and uncertain how their inputs can be effectively fostered; Ralph can use this phase to encourage adaptability and set expectations for team to get a sense of what is expected

Storming, at this stage the team begins to openly challenge each other and conflict begins to arise; Ralph must understand team dynamics are emerging and guiding the team through this phase lets Ralph know that the team is aware/ and know what is expected after this

Norming, the teams figure out how to set aside differences and work together; Ralph can use this phase to foster a collaborative culture and initiate a set of standards that must be met so that the team does not lose their collaborative culture

Performing- The team collaborates and effectively reaches their optimal level of performance and can work together through different situations/circumstances. Ralph can support the team in constantly growing and becoming more adaptable/flexible to deliver more effectively.

Question 3

Pair programming

Main strengths/benefits	Discussion
Better designs & fewer bugs Continuous review	By having programmers observe each other's work, Cisco solutions can develop better software quality and achieve designs that meet customer expectations effectively as programmers can guide, help each other to achieve high quality code.
Defects are caught early	As the code is being developed, testing must occur and defects are caught early allowing Cisco solutions to spend less time on catching/correcting defects after product has been developed and help them meet deadlines
Increased productivity rather than cost	What is achieved by a second developer is greater and contributes more value to customers/business which Cisco can prioritize to increase customer satisfaction and should overlook this cost.
Learning & training	As the development progresses constant sharing of methods/knowledge can help Cisco programmers develop/learn each other's skills so both programmers can be more effective/work faster and ensure customer value is always delivered and on time.
Overcoming difficult problems	Pairs are more effective in solving complex problems; Cisco solutions can adapt to this approach to navigate large amounts of time spent on solving problems individually and effectively meet deadlines.

Question 4

Managing communications with offshore teams in Agile

geographic separation – not being able to meet team members physically in person(f2f) can lead to misunderstandings and misinterpretations which contribute to unclear documentation and in accurate requirements definition causing a project to fail in meeting customer objectives.

Cultural and language differences – Teams in geographic locations speak different languages and this language barriers can hinder effective communication which can lead to dissatisfaction in delivering a product effectively and hinder team spirit/motivation of getting a sprint done.

time zone differences – workers in another geographic location will find it tough to agree on a time zone that workers can easily accommodate to participate and host meetings; scrum meetings will need to be strategized in overlapping hours to happen every day.

Technical dependance – without technology no productivity can be completed, and this can be detrimental as any technological issues can halt productivity and communication. E.g. if a power outage occurs communication cannot take place at all and this can be detrimental in delivering projects on deadlines

Team collaboration challenges – teams aren't able to effectively communicate and collaborate as they are in different locations this can hinder trust and discourage team collaboration

Question 5

5 Characteristics of complex adaptive systems (CAS)

- Emergent, CAS come into being (interactions happen randomly) rather than being planned
- Sub-optimal, the focus is not always on delivering a perfect solution but rather on delivering MVP (minimum viable product) which offers business value.
- Requisite variety, thrive on variety, agile teams are made up of people with different perspectives, skills and competencies.
- Connectivity, agents in complex adaptive systems stay connected to each other, connectivity must not be forced collaborating and working together initiates the path to the success of agile teams
- Simple rules, the rules governing the system should not be complex and should be simple enough for everyone to understand.

Question 6

6.1) Product vision

For (target customer)	For Lecturers & students at Boston City College
Why (statement of need/opportunity)	Students seek centralized access to resources and collaboration within the same qualification, while lecturers require oversight to ensure productive engagement.
What(product/project) name	A digital learning platform (LMS) where students can just login to their respective departments that provides a collaborative learning hub
So that (key benefit)	So that students can access resources, connect with students within the same qualification, share knowledge, and work together on assessments while lecturers can monitor student activity.
Unlike (primary/competitive alternative)	Unlike using communication tools like WhatsApp, which can be inefficient in monitoring all students, and using a basic platform lacking collaboration tools for students in the same qualification
Our product (Statement of primary differentiation)	Our product bridges the need for collaboration with students within the same qualification and ensures productive engagement through monitoring and fostering a supportive learning environment.

6.2) The five steps of the Theory of Constraints (TOC)

1. Identify: Identify bottlenecks (e.g., pages take long to display info, team delays in developing system) in Boston City's digital platform (LMS). Developers can focus on one issue at a time.
2. Exploit: Optimize constraints (e.g., improve load speed, prioritizing essential/functionality tasks). Developers create solutions to support system functionality, improving overall flow.
3. Subordinate: Ensure constraint is optimized by other processes and solutions doesn't cause issues elsewhere. Developers test to prevent bottlenecks from affecting other functionalities.
4. Elevate: Eliminate bottlenecks (e.g., upgrade infrastructure, assign new development team). System runs smoothly without the same bottleneck.

5. Check for new constraints: Continuously test and monitor until all bottlenecks have been completely addressed to ensure platform remains efficient and to prevent piling of constraints that can derive system functionality.

6.3) Conducting an Exploratory 360 Assessment helps IT students plan Boston City College's digital platform by:

- Identifying required functionality, stakeholders, user stories, and personas to understand expectations
- Determining if the project adds value and is deliverable with available resources
- Enabling IT teams to plan and prioritize achieving value